

QUALITY ASSURANCE PROJECT PORTFOLIO

Employee Leave Management System (ELMS)

Your Name

Quality Assurance Engineer

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Tools: JIRA | Selenium | Postman | SQL | TestRail

Project Overview

The Employee Leave Management System (ELMS) is a web-based application designed to manage employee leave requests, approval workflows, leave balance tracking, and administrative reporting.

As the QA Engineer on this project, I was responsible for ensuring functional accuracy, workflow validation, data consistency, and overall system reliability prior to production release.

Responsibilities

- Analyzed Business Requirement Documents (BRD) and Functional Requirement Specifications (FRS)
- Designed detailed test scenarios and test cases
- Performed manual functional and regression testing
- Logged and tracked defects using JIRA
- Conducted API validation using Postman
- Performed database validation using SQL queries
- Coordinated with developers for defect resolution
- Prepared test execution and summary reports

Test Strategy & Approach

Testing was conducted using a structured QA process:

1. Requirement Analysis
2. Test Planning
3. Test Case Design
4. Test Execution
5. Defect Reporting & Retesting
6. Regression Testing
7. Final Sign-Off

Testing Types Performed:

- Functional Testing
- Regression Testing
- UI/UX Testing
- Negative Testing
- Boundary Value Testing
- API Testing

Key Modules Tested

1. User Authentication (Login/Logout)
2. Leave Application Submission
3. Leave Balance Calculation
4. Manager Approval/Rejection Workflow
5. Leave History Tracking
6. Admin Dashboard & Reporting
7. Role-Based Access Control

Sample Test Cases

Test Case ID: TC_ELMS_001

Title: Verify Employee Login with Valid Credentials

Precondition: User account exists

Steps:

1. Enter valid username
2. Enter valid password
3. Click Login

Expected Result: User redirected to employee dashboard

Status: Passed

Test Case ID: TC_ELMS_005

Title: Validate Leave Balance Deduction After Approval

Steps:

1. Employee applies for 3 days leave
2. Manager approves request
3. Navigate to Leave Balance

Expected Result: Leave balance reduced by 3 days

Status: Passed

Sample Defect Report

Bug ID: ELMS_BUG_014

Title: Leave Balance Not Updating After Manager Approval

Severity: High

Priority: Critical

Environment: Chrome Browser / Windows 11

Steps to Reproduce:

1. Submit leave request
2. Approve request as manager
3. Check leave balance

Expected Result: Leave balance updated correctly

Actual Result: Balance remains unchanged

Status: Fixed and Successfully Retested

Test Execution Summary

Total Test Cases Designed: 45

Total Executed: 45

Passed: 42

Failed: 2

Blocked: 1

Defects Identified: 18

Critical: 3

High: 5

Medium: 7

Low: 3

Final Status: System stable after fixes. Recommended for production deployment.

Automation Contribution (If Applicable)

Automated Regression Scenarios:

- Login Validation
- Leave Submission Workflow
- Leave Approval Process

Framework Used:

- Selenium with Java
- TestNG
- Page Object Model (POM)

Impact:

Reduced regression execution time by 55% and improved release confidence.

Value Delivered to Client

- Improved system reliability before launch
- Prevented critical payroll-related leave miscalculations
- Ensured accurate leave balance tracking
- Reduced post-release production defects
- Delivered structured QA documentation for audit compliance